

Machine Learning for teens

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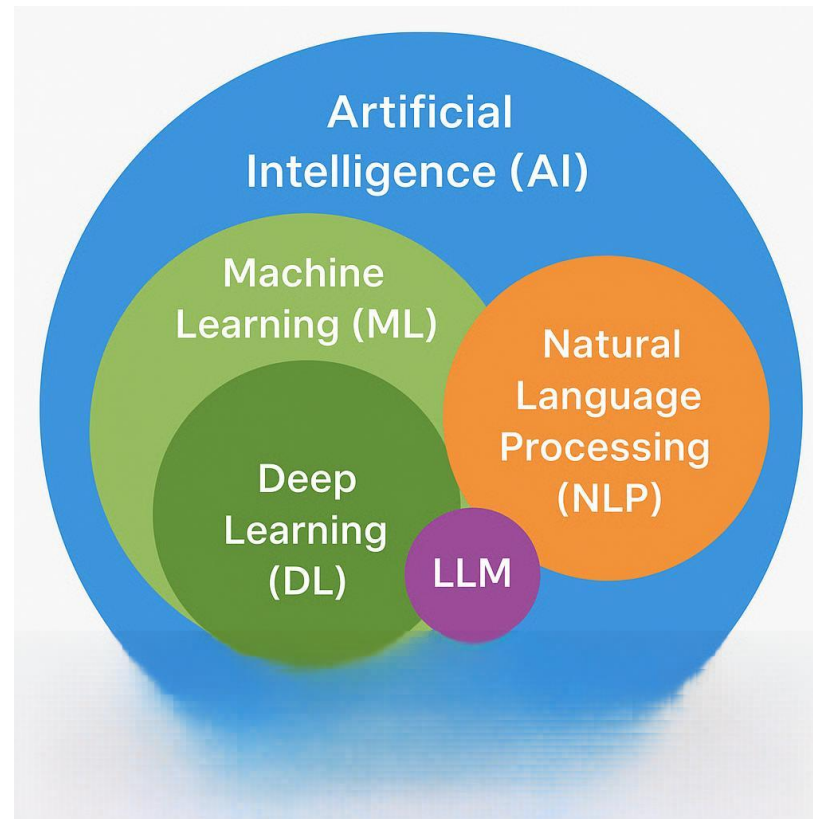
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What is AI

- ▶ AI is like giving computers the ability to think, learn, and make decisions like humans
 - ▶ Smart Assistants (like Siri or Copilot)
 - ▶ Shopping Recommendations

AI-ML-NLP and LLM



Types of learning

1. Supervised learning

Learning from labeled data (input + correct output).

2. Unsupervised learning

Learning from unlabeled data (no correct answers).

goal: Find hidden patterns or groupings.

3. Reinforcement Learning

Learning by trial and error through rewards and penalties.

example: Robot from point A to point B

Types of problem by outcome or target variable

- ▶ Discrete : have classification task
- ▶ Continuous : have regression task

Jupyter notebook

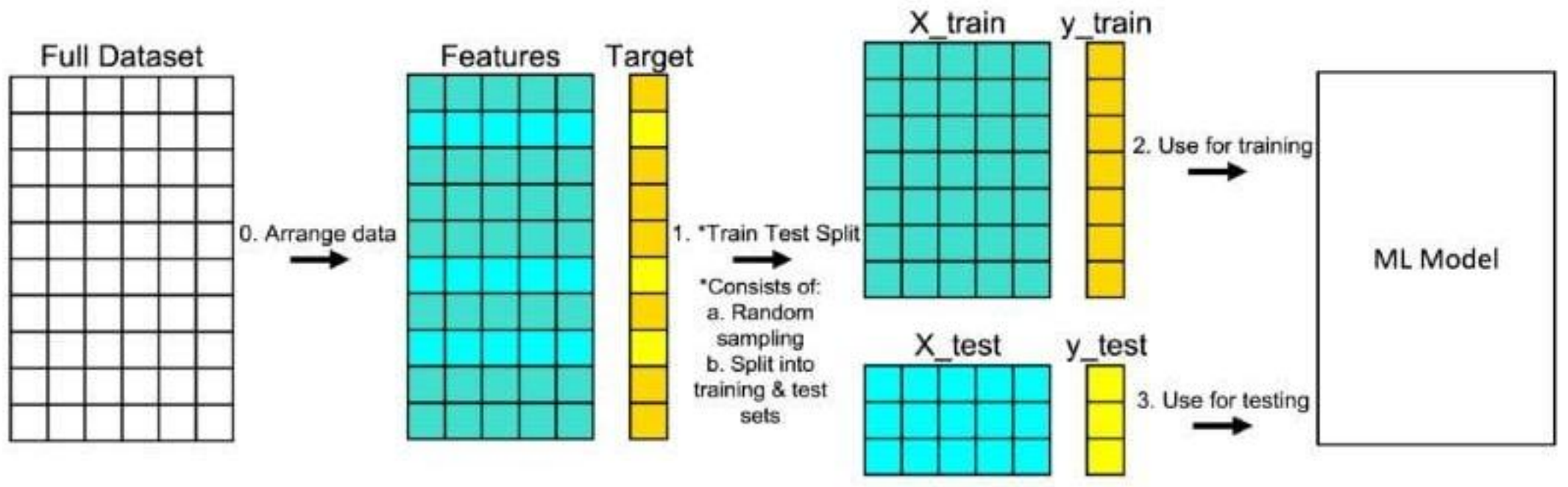
- ▶ Its an IDE similar to vscode
- ▶ Install by this
 - ▶ `pip install notebook`
- ▶ Run it
 - ▶ `jupyter notebook`

Target variable and Features

- ▶ Target variable is the column we try to predict
- ▶ Features are other columns

Data splitting

- ▶ We need to do that because model should predict the data that never seen. Usually, we split data into train and test . (75 to 25 is default)
- ▶ Training data – where the model learns
- ▶ Test data – where we check how well it learned

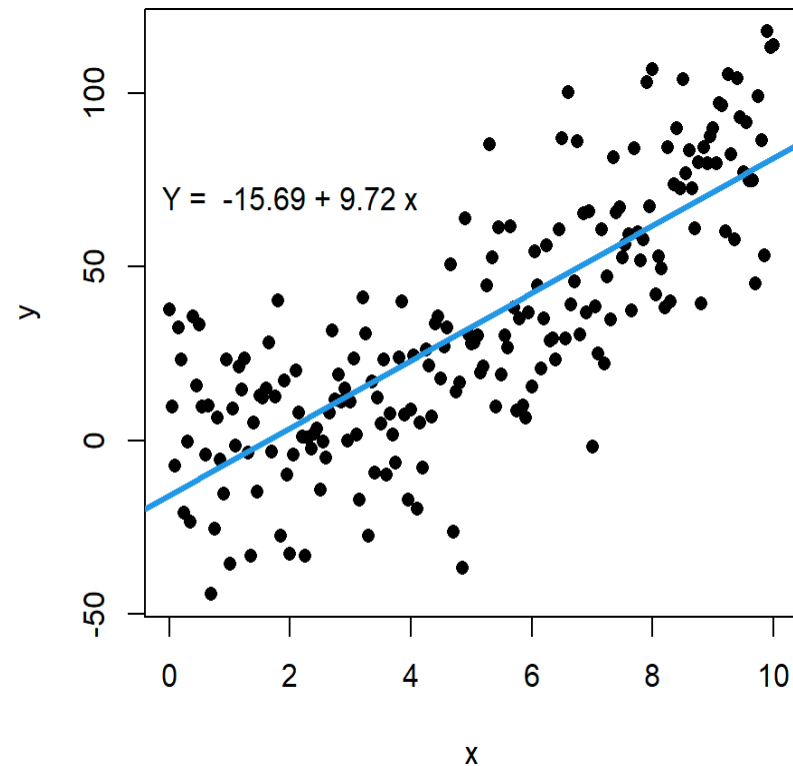


Data split ML vs DL

- ▶ Standard ML Data Split
 - ▶ Full dataset
 - ▶ |—— Training set (e.g. 80%)
 - ▶ |—— Test set (e.g. 20%)
- ▶ Standard in Deep Learning and fine-tuning
 - ▶ Full dataset
 - ▶ |—— Training set → model learns
 - ▶ |—— Validation set → model tunes itself
 - ▶ |—— Test set → final evaluation

What is Linear Regression

- Linear regression try to find a linear relationship between features and target variable. now lets say x represent feature column and y target the equation between them is $y=mx+b$



several feature columns

- ▶ What if we can several feature columns?
 - ▶ What is the equation?
 - ▶ $y=b+w_1x_1+w_2x_2+w_3x_3+\dots+w_nx_n$
 - ▶ Where $x_1 \dots x_n$ are feature columns
 - ▶ $w_1 \dots w_n$ coefficients
 - ▶ b is intercept

Linear regression

- ▶ Prerequisites
 - ▶ Read about pandas, sklearn , matplotlib and seaborn
- ▶ Imagine you are trying to predict price of a home based on its area
 - ▶ Download home_price.csv. the [link](#)
 - ▶ Try to load the data and then plot it
 - ▶ Make linear model
 - ▶ Do the prediction
 - ▶ Calculate r2 score
 - ▶ If your model is great $\rightarrow R^2$ is close to 1
 - ▶ If your model is weak $\rightarrow R^2$ is close to 0
 - ▶ If your model is terrible $\rightarrow R^2$ is negative

homework

****Load the Dataset****

Import the dataset into a pandas DataFrame. Display the first few rows using `.head()`.

****Explore the Data****

Use `.info()` and `.describe()` to understand the structure. Check for missing values and data types.

****Visualize Relationships****

Plot scatter plots between `MedHouseVal` and each feature.

Look at how each feature changes with `MedHouseVal` to find which ones might be useful for prediction.

****split data into train and test**** for taring

****now use train the model****

****find R2 score****

****is the result good enough??？****

Any question

